KodeToCareer

INDUSTRY-READY PROGRAM 2025

Complete Industry-Ready MLOps & LLMOps Roadmap

Become a Production AI Engineer in 6 Months

- 28+ Weeks Curriculum
- 20+ Modules
- 15+ Production Projects
- Weekly Labs
- Capstone Project
- Cloud + K8s + LLMOps
- Job Ready

ABOUT THE PROGRAM

Your Path to Becoming a Production AI Engineer

6	5	28+	15+
Months Duration	Days Per Week	Weeks of Learning	Production Projects

This intensive program transforms you into a production-ready AI engineer capable of building, deploying, scaling, and monitoring machine learning systems at enterprise scale. You will gain hands-on experience with the exact tools, platforms, and practices used by teams at Google, NVIDIA, OpenAI, Microsoft, and Databricks. Every module is designed around real-world production scenarios, ensuring you graduate with a portfolio of deployable systems.

Graduates of this program are equipped to step into high-impact roles across the AI industry, from MLOps and LLMOps engineering to AI infrastructure and platform engineering. The curriculum covers the entire AI production lifecycle, from foundational Python and Linux skills through advanced GPU optimization and system design.

M

MLOps Engineer

Design and manage end-to-end ML pipelines, from model training to production deployment, ensuring reliability, scalability, and continuous integration of machine learning systems.

I

AI Infrastructure Engineer

Build and maintain the computational infrastructure that powers AI workloads, including GPU clusters, Kubernetes environments, and cloud-based ML platforms.

P

ML Platform Engineer

Develop internal ML platforms and tools that enable data scientists and ML engineers to build, train, and deploy models efficiently at scale.

L

LLMOps Engineer

Specialize in the operational lifecycle of large language models, including fine-tuning, serving, monitoring, and optimizing LLM-based applications in production.

D

ML Deployment Engineer

Focus on the deployment and serving of ML models, implementing CI/CD pipelines, model versioning, A/B testing, and canary release strategies.

S

AI Systems Engineer

Design and architect complex AI systems that integrate multiple ML models, data pipelines, and infrastructure components into cohesive production solutions.

LEARNING JOURNEY

22-Step Progression Roadmap

Your journey follows a carefully sequenced 22-step progression, each building on the previous. You start with foundational skills and progressively advance through machine learning, deep learning, containerization, orchestration, cloud platforms, and specialized MLOps and LLMOps practices until you reach system design mastery and career readiness.

AI Foundations — Python — Linux — Git — Machine Learning — Deep Learning — Docker — Kubernetes — Cloud

— CI/CD — ML Pipelines — Data Engineering — Model Serving — LLMOps — GPU Infra — Optimization — Monitoring —

DevSecOps — Troubleshooting — System Design — Career Prep — Capstone

CURRICULUM

21 Modules. Complete Industry Coverage.

Every module is production-focused, designed around real-world AI engineering challenges. Each includes hands-on exercises, lab work, and where applicable, a production-level mini project. The curriculum progresses from foundational skills to advanced system design, ensuring no gaps in your knowledge.

MODULE 00 — 1 weeks

AI Foundations

- What is AI, ML, and Deep Learning
- AI Industry Landscape & Career Paths
- Mathematics for AI (Linear Algebra, Calculus, Statistics)
- AI Development Lifecycle Overview
- Introduction to MLOps & LLMOps
- Setting Up AI Development Environment

MODULE 01 — 1.5 weeks

Python for AI Engineering

- Python Fundamentals & Advanced Concepts
- Object-Oriented Programming
- File Handling, Error Management
- Virtual Environments & Package Management
- APIs (REST, WebSockets)
- Asynchronous Programming

MINI PROJECT
Build Python CLI Tool for ML Workflow

MODULE 02

1 weeks

Linux for AI & Cloud

- Linux File System & Navigation
- Shell Scripting & Automation
- Process Management & System Monitoring
- Networking Fundamentals
- SSH, SCP, and Remote Access
- Package Management (apt, yum)

MODULE 03

1 weeks

Git & Collaborative Development

- Git Fundamentals (Commit, Branch, Merge)
- Advanced Git (Rebase, Cherry-pick, Stash)
- Pull Requests & Code Reviews
- Git Flow & Trunk-Based Development
- GitHub/GitLab CI Integration
- Version Control Best Practices

MODULE 04

2 weeks

Machine Learning Fundamentals

- Supervised Learning (Regression, Classification)
- Unsupervised Learning (Clustering, Dimensionality Reduction)
- Model Evaluation & Validation (Cross-Validation, Metrics)
- Feature Engineering & Selection
- Scikit-learn Pipeline API
- Hyperparameter Tuning (Grid Search, Random Search)
- Ensemble Methods (Random Forest, Gradient Boosting)
- ML Model Interpretability (SHAP, LIME)

MINI PROJECT
Fraud Detection System

MODULE 05

2 weeks

Deep Learning

- Neural Network Fundamentals
- PyTorch & TensorFlow Frameworks
- Convolutional Neural Networks (CNNs)
- Recurrent Neural Networks (RNNs, LSTMs)
- Transfer Learning & Fine-Tuning
- Model Training & Optimization
- GPU Acceleration Basics

MINI PROJECT
Image Classification API

MODULE 06

1.5 weeks

Docker & Containerization

- Docker Fundamentals (Images, Containers, Volumes)
- Dockerfile Best Practices for ML
- Multi-Stage Builds for ML Workloads
- Docker Compose for Multi-Container Apps
- Container Registry Management
- Docker Networking & Storage

MINI PROJECT
Containerized ML APIs

MODULE 07

2 weeks

Kubernetes Orchestration

- Kubernetes Architecture (Control Plane, Nodes)
- Pods, Deployments, Services, Ingress
- ConfigMaps, Secrets, and Volumes
- Helm Charts & Package Management
- Auto-Scaling (HPA, VPA, Cluster Autoscaler)
- Resource Management & Limits
- Kubernetes Networking & Security
- Troubleshooting K8s Clusters

MINI PROJECT
Kubernetes Deployment

MODULE 08

1.5 weeks

Cloud Platforms (AWS)

- AWS Core Services (EC2, S3, IAM, VPC)
- AWS SageMaker for ML Workloads
- EKS (Elastic Kubernetes Service)
- ECR (Elastic Container Registry)
- CloudWatch Monitoring & Logging
- IAM Roles & Security Policies
- Cost Optimization Strategies

MINI PROJECT
Cloud ML Deployment

MODULE 09

1.5 weeks

CI/CD for ML

- CI/CD Fundamentals & ML-Specific Challenges
- GitHub Actions / GitLab CI for ML
- Automated Testing for ML Code
- Model Registry & Versioning
- Automated Model Training Pipelines
- Blue-Green & Canary Deployments
- Rollback Strategies for ML Models

MINI PROJECT
End-to-End ML Pipeline

MODULE 10

2 weeks

ML Pipeline Engineering

- MLflow (Experiments, Models, Registry)
- Kubeflow Pipelines Architecture
- Feature Stores (Feast, Tecton)
- Data Versioning (DVC)
- Pipeline Orchestration (Airflow, Prefect)
- Pipeline Monitoring & Alerting
- Production Pipeline Patterns

MINI PROJECT
Production ML API

MODULE 11

1.5 weeks

Data Engineering for ML

- ETL/ELT Pipelines
- Apache Kafka for Real-Time Data
- Apache Spark for Large-Scale Processing
- Data Warehousing (Redshift, BigQuery)
- Data Quality & Validation
- Streaming vs Batch Processing
- Schema Evolution & Management

MODULE 12

1.5 weeks

Model Serving & Inference

- Model Serving Patterns (REST, gRPC, BentoML)
- FastAPI for ML Model APIs
- A/B Testing for ML Models
- Model Versioning & Traffic Splitting
- Batch vs Real-Time Inference
- API Gateway & Load Balancing
- Caching Strategies for ML Inference

MINI PROJECT
Production RAG System

MODULE 13

2 weeks

LLMOps & Production LLMs

- LLM Architecture & Tokenization
- Prompt Engineering & Management
- RAG (Retrieval-Augmented Generation)
- LangChain, LangGraph, LlamaIndex
- Vector Databases (FAISS, Qdrant, ChromaDB)
- LLM Evaluation & Benchmarking
- LLM Cost Optimization
- Production LLM Architecture Patterns

MINI PROJECT
Optimize LLM Inference

MODULE 14

1.5 weeks

GPU Infrastructure

- GPU Architecture (NVIDIA A100, H100)
- Multi-GPU & Distributed Training
- GPU Scheduling in Kubernetes
- NVIDIA GPU Operator & Device Plugins
- MIG (Multi-Instance GPU) Configuration
- GPU Monitoring & Utilization
- Cost-Effective GPU Strategies

MINI PROJECT
Model Monitoring

MODULE 15

1.5 weeks

Model Optimization

- Model Quantization (INT8, FP16)
- Knowledge Distillation
- Pruning & Sparsity
- vLLM & TensorRT for Fast Inference
- NVIDIA Triton Inference Server
- Model Compilation (ONNX, TensorRT)
- Benchmarking & Performance Profiling

MODULE 16

1.5 weeks

ML Monitoring & Observability

- Prometheus & Grafana for ML
- Model Drift Detection (Data, Concept, Prediction)
- Custom ML Metrics & Dashboards
- Alerting & Incident Response
- Log Aggregation (ELK, Loki)
- Distributed Tracing (Jaeger, OpenTelemetry)
- SLIs, SLOs, and Error Budgets for ML

MODULE 17

1 weeks

DevSecOps for ML

- Security Fundamentals for ML Systems
- Container Security Scanning
- Secret Management (Vault, K8s Secrets)
- Network Policies & RBAC
- Compliance & Audit Logging
- Vulnerability Management
- Secure ML Pipeline Design

MODULE 18

1.5 weeks

Production Troubleshooting

- Debugging Production ML Systems
- Performance Bottleneck Analysis
- Memory & Resource Profiling
- Log Analysis & Root Cause Identification
- Chaos Engineering for ML
- Incident Management & Post-Mortems
- Common Production Failure Patterns

MODULE 19

1.5 weeks

AI System Design

- Designing Scalable ML Architectures
- Microservices for ML Systems
- Event-Driven Architecture for ML
- Data Pipeline Architecture
- Cost-Performance Trade-offs
- Multi-Tenant ML Platforms
- System Design Interviews & Case Studies

MODULE 20

2 weeks

Career Preparation & Capstone

- Resume Building for AI/ML Roles
- LinkedIn & Portfolio Optimization
- Interview Preparation (Technical + Behavioral)
- System Design Interview Practice
- Salary Negotiation Strategies
- Industry Networking & Community
- Final Capstone Project Execution

MINI PROJECT
Final Capstone Project

PROJECTS

15+ Production Projects

Every project in this program mirrors real production systems at leading AI companies. You will build, containerize, deploy, and monitor complete ML and LLM systems, graduating with a portfolio that demonstrates production-level engineering capability.

Fraud Detection System

Build an end-to-end fraud detection system with model training, API serving, and monitoring dashboard.

Python | Scikit-learn | FastAPI | Docker

Image Classification API

Deploy a deep learning image classification model as a production REST API with versioning.

PyTorch | FastAPI | Docker | MLflow

Containerized ML APIs

Containerize multiple ML models with Docker Compose and deploy with health checks.

Docker | Docker Compose | FastAPI | Python

Kubernetes Deployment

Deploy ML workloads on Kubernetes with auto-scaling, monitoring, and rolling updates.

Kubernetes | Helm | Prometheus | Grafana

Cloud ML Deployment

Deploy ML pipeline on AWS with SageMaker, EKS, and CloudWatch integration.

AWS | SageMaker | EKS | CloudWatch

End-to-End ML Pipeline

Build a complete CI/CD pipeline for ML with automated training, testing, and deployment.

GitHub Actions | MLflow | Docker | Kubernetes

Production ML API

Create a production-grade ML API with MLflow model registry, feature store, and pipeline orchestration.

MLflow | Kubeflow | FastAPI | Airflow

Production RAG System

Build a production RAG system with vector database, embedding management, and LLM serving.

LangChain | FAISS | FastAPI | Docker

Optimize LLM Inference

Optimize LLM inference with quantization, vLLM, and TensorRT for production throughput.

vLLM | TensorRT | NVIDIA Triton | PyTorch

Model Monitoring

Implement comprehensive model monitoring with drift detection, alerting, and dashboards.

Prometheus | Grafana | Python | Kafka

WEEKLY LABS

Hands-On Production Labs

Weekly production labs provide intensive, hands-on experience with the exact challenges you will face in production environments. Each lab is designed to simulate real-world scenarios and build muscle memory for critical operations.

LAB

Build ML Pipelines

Construct end-to-end ML training and deployment pipelines with MLflow and Kubeflow.

LAB

Kubernetes Deployment

Deploy and scale containerized ML workloads on production Kubernetes clusters.

LAB

GPU Optimization

Configure GPU scheduling, multi-GPU training, and optimize inference throughput.

LAB

Rollbacks & Recovery

Implement blue-green deployments, canary releases, and automated rollback strategies.

LAB

Drift Detection

Set up data drift, concept drift, and prediction drift monitoring with alerting.

LAB

Debugging Production

Diagnose and fix real production issues including memory leaks and latency spikes.

LAB

Scaling Systems

Auto-scale ML serving infrastructure based on request load and GPU utilization.

LAB

Benchmarking

Measure and optimize model inference latency, throughput, and cost per prediction.

LAB

Chaos Testing

Introduce controlled failures to test system resilience and recovery procedures.

CAPSTONE

Final Capstone Project

The capstone project is your culminating achievement, demonstrating mastery across the entire MLOps and LLMOps stack. Choose from four production-grade options, each representing a complete, deployable system that showcases your ability to design, build, and operate AI infrastructure at enterprise scale.

CAPSTONE

Production Fraud Detection Platform

Design and deploy a real-time fraud detection platform with model training pipelines, Kubernetes deployment, monitoring dashboards, and automated retraining triggers. Full production stack with CI/CD, security, and observability.

Python · Scikit-learn · FastAPI · Docker · Kubernetes · AWS · Prometheus · Grafana · MLflow · GitHub Actions

CAPSTONE

Enterprise LLM Inference Cluster

Set up a production LLM inference cluster on Kubernetes with GPU scheduling, model optimization (quantization, TensorRT), load balancing, auto-scaling, and comprehensive monitoring. Optimize for throughput and cost efficiency.

Kubernetes · NVIDIA GPU Operator · vLLM · TensorRT · Triton · Prometheus · Grafana · Helm · PyTorch

CAPSTONE

Production RAG System

Build an enterprise-grade RAG system with document processing pipelines, vector databases, embedding management, LLM orchestration, caching layers, and production monitoring. Include A/B testing and evaluation frameworks.

LangChain · LangGraph · LlamaIndex · FAISS · Qdrant · FastAPI · Docker · Kubernetes · MLflow

CAPSTONE

Real-Time Recommendation System

Design a real-time recommendation system with streaming data pipelines (Kafka), feature stores, model serving infrastructure, A/B testing, and production monitoring. Deploy on Kubernetes with auto-scaling.

Apache Kafka · Apache Spark · Feast · FastAPI · Docker · Kubernetes · AWS · Prometheus · Grafana

TECHNOLOGIES

Production Technology Stack

Master the exact tools and platforms used by AI engineering teams at the world's leading technology companies. Our technology stack is continuously updated to reflect current industry standards and emerging best practices.

PROGRAMMING

Python · SQL · Bash/Shell · YAML · Terraform

MACHINE LEARNING

Scikit-learn · XGBoost · LightGBM · Pandas · NumPy

DEEP LEARNING

PyTorch · TensorFlow · Hugging Face · Transformers

DATA ENGINEERING

Apache Kafka · Apache Spark · Airflow · DVC · Feast

MLOPS

MLflow · Kubeflow · DVC · Weights & Biases

LLMOPS

LangChain · LangGraph · LlamaIndex · vLLM

DEPLOYMENT

FastAPI · Docker · Kubernetes · Helm · BentoML

CLOUD

AWS · SageMaker · EKS · ECR · CloudWatch

INFRASTRUCTURE

Terraform · Ansible · Nginx · Envoy

OBSERVABILITY

Prometheus · Grafana · Jaeger · ELK Stack

GPU & OPTIMIZATION

CUDA · TensorRT · NVIDIA Triton · ONNX

SECURITY

Vault · OPA · Trivy · Falco

SKILLS

Skills You'll Master

Upon completing this program, you will have demonstrable proficiency in 30+ critical AI engineering skills, spanning programming, machine learning, infrastructure, deployment, observability, and security. These skills are validated through hands-on projects, weekly labs, and your final capstone.

Python SQL Linux Git Docker Kubernetes MLflow Kubeflow FastAPI AWS PyTorch TensorFlow LangChain LangGraph LlamaIndex FAISS Qdrant ChromaDB vLLM Triton Prometheus Grafana Kafka Spark CI/CD GPU Optimization AI System Design Production Troubleshooting Monitoring Security

WHY THIS PROGRAM

Why Choose This Program

This program stands apart from generic AI courses through its relentless focus on production readiness. Every module, project, and lab is designed to build the exact competencies that hiring managers at top AI companies are looking for. Here are twelve reasons why this program delivers unmatched value.

Industry-Ready Curriculum
Designed with input from hiring managers at top AI companies, covering exactly what production teams need.

Production-Level Projects
15+ real-world projects that mirror actual production systems, not toy examples.

Hands-on Kubernetes
Deep Kubernetes expertise with GPU scheduling, auto-scaling, and Helm chart development.

Cloud Deployment
Real AWS deployment experience with SageMaker, EKS, and production-grade infrastructure.

LLMOps Specialization
Comprehensive LLMOps coverage including RAG, vector databases, and LLM optimization.

Real GPU Infrastructure
Work with actual NVIDIA GPUs, learn scheduling, optimization, and distributed training.

Weekly Labs
Hands-on weekly production labs covering Kubernetes, GPU optimization, and chaos testing.

CI/CD Mastery
End-to-end CI/CD pipelines specifically designed for ML workflows and model deployment.

Monitoring & Observability
Production monitoring with Prometheus, Grafana, drift detection, and alerting systems.

System Design
AI system design interviews and case studies from real production architectures.

Career Preparation
Resume building, interview prep, LinkedIn optimization, and salary negotiation strategies.

Interview Preparation
Technical and behavioral interview practice with mock sessions and feedback.

CAREERS

Career Opportunities

The demand for MLOps and LLMOps engineers has never been higher. Companies across every industry are racing to deploy AI systems in production, creating unprecedented career opportunities for engineers who can bridge the gap between model development and production infrastructure.

MLOps Engineer $120K-$**200K+** Very High Demand	**AI Infrastructure Engineer** $130K-$**210K+** Very High Demand	**ML Platform Engineer** $125K-$**205K+** High Demand
LLMOps Engineer $140K-$**230K+** Extremely High Demand	**AI Systems Engineer** $135K-$**220K+** Very High Demand	**ML Deployment Engineer** $115K-$**195K+** High Demand

According to industry reports, the global MLOps market is projected to grow from $1.3billionin2024toover$12 billion by 2030, reflecting a compound annual growth rate exceeding 40%. This explosive growth is driven by enterprise AI adoption across fintech, healthcare, e-commerce, autonomous vehicles, and generative AI applications. Companies like Google, Meta, Netflix, Uber, and Stripe are actively hiring engineers who can deploy, monitor, and scale ML systems in production.

The talent gap in MLOps and LLMOps is estimated at over 300,000 unfilled positions globally. Unlike traditional software engineering roles, these positions require a rare combination of machine learning expertise, infrastructure knowledge, and DevOps skills, making qualified professionals highly sought after and commanding premium compensation packages.

Graduates of this program have gone on to work at leading technology companies, AI startups, and enterprise organizations. The combination of production projects, weekly labs, and a comprehensive capstone ensures that you enter the job market with demonstrable, portfolio-ready experience that sets you apart from candidates with only theoretical knowledge or course-based certifications.

KodeToCareer

Build • Deploy • Scale • Optimize • Monitor

AI Systems Like Industry Professionals

Enroll Today & Become a Production AI Engineer

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